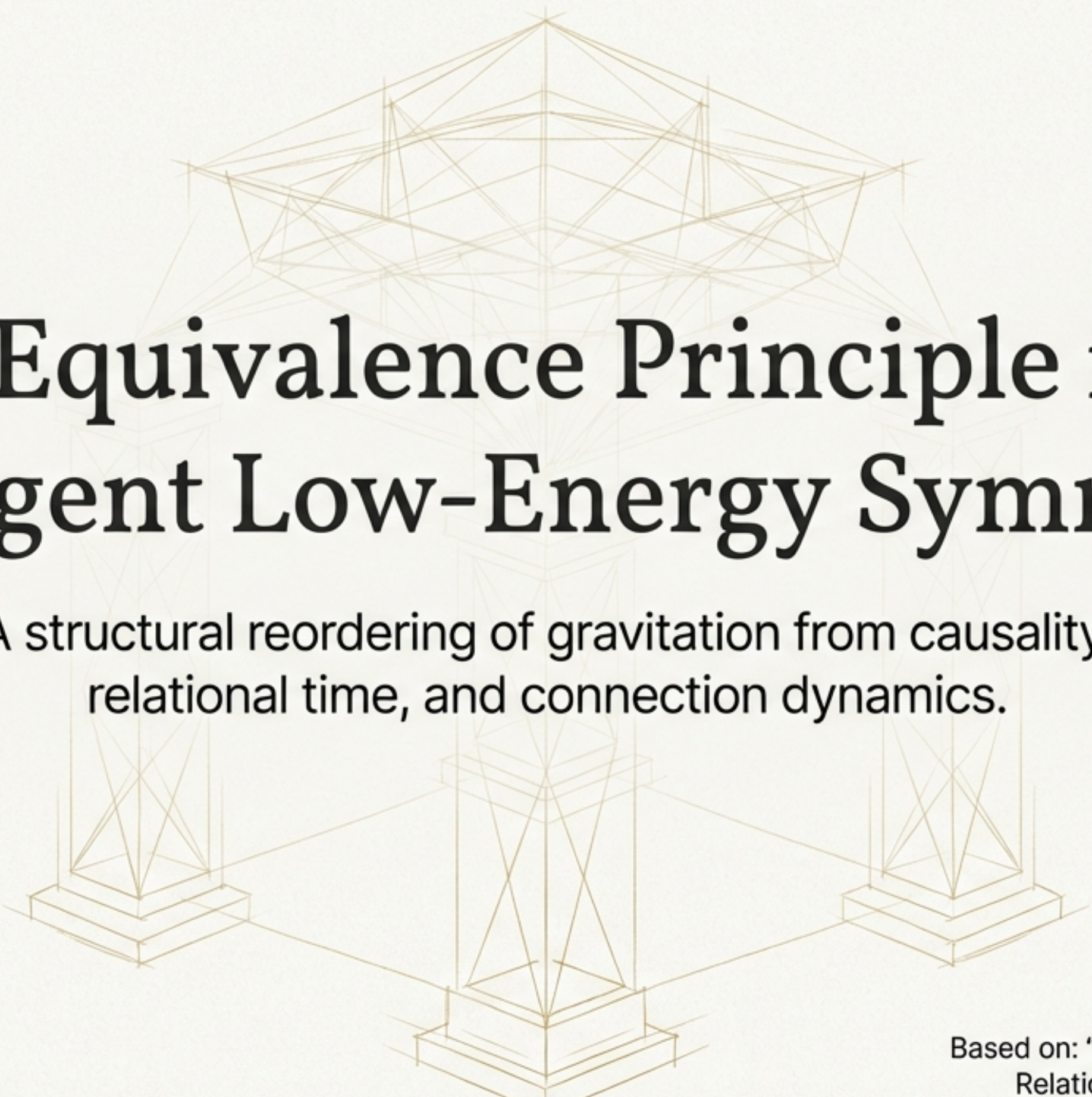




The Equivalence Principle is an Emergent Low-Energy Symmetry

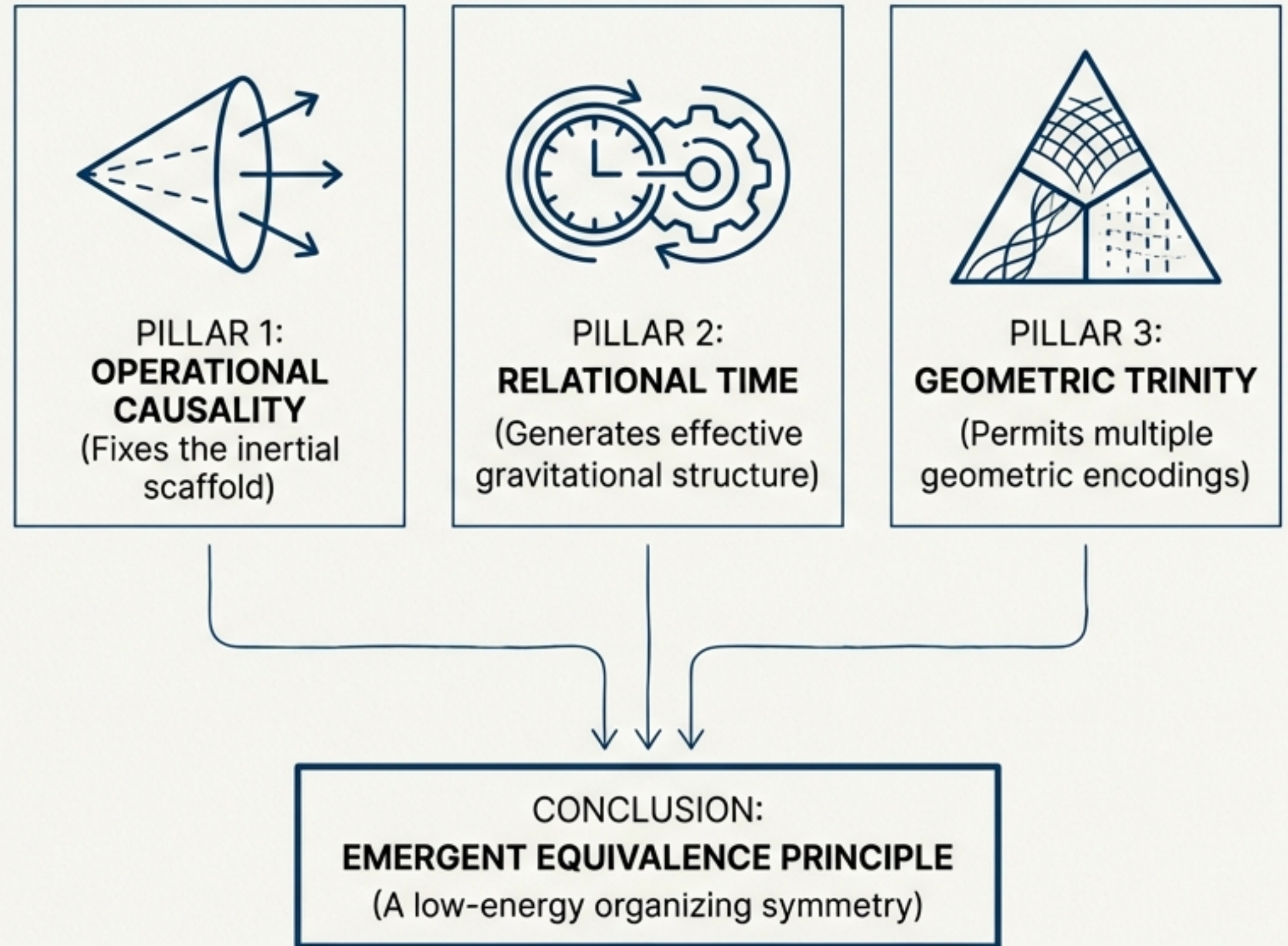
A structural reordering of gravitation from causality, relational time, and connection dynamics.

Omar Iskandarani
Based on: "Emergent Equivalence Principle from
Relational Time and Connection Dynamics"

The Argument: Reordering Logic, Not Replacing Phenomenology

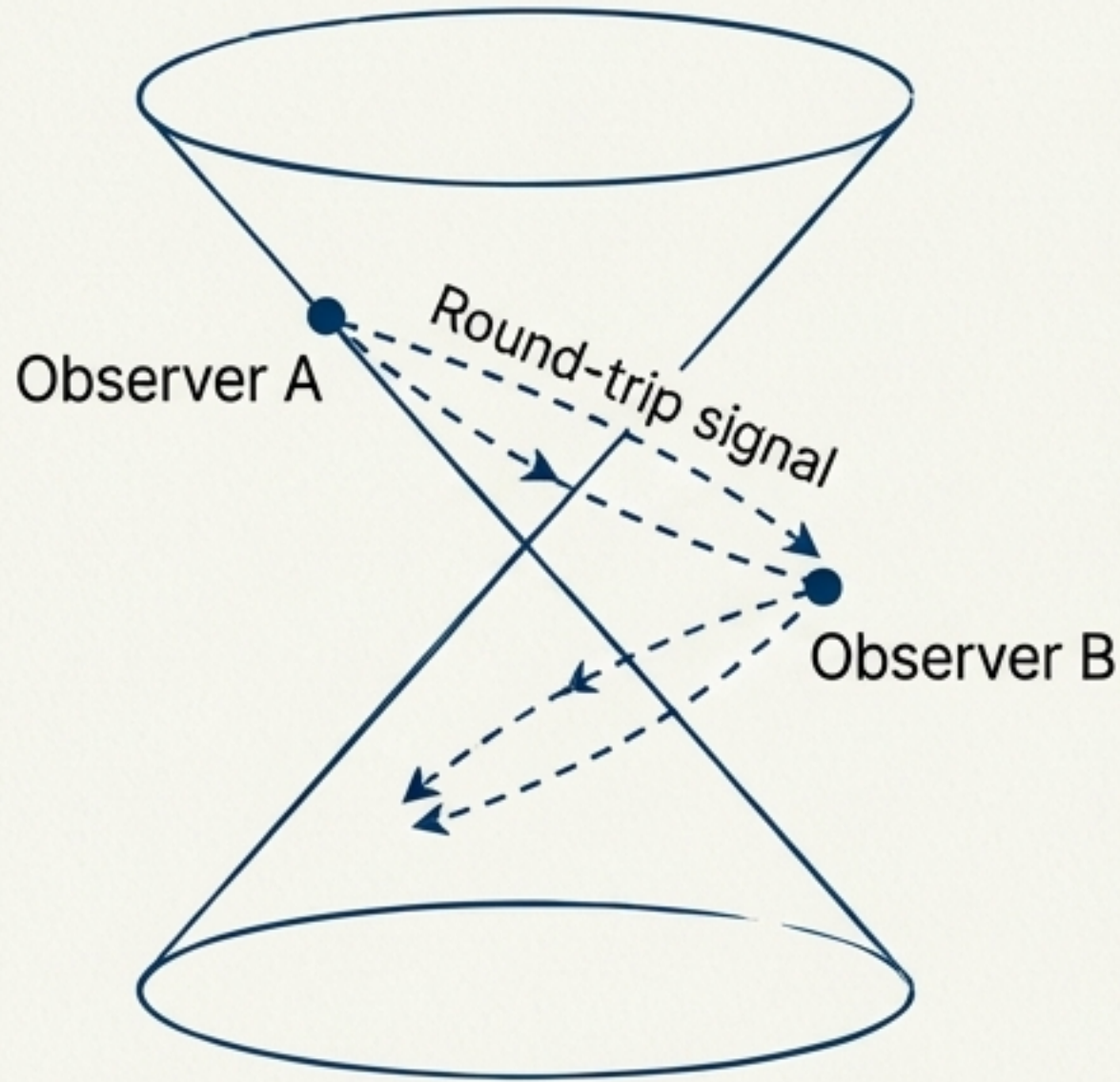
Established physics remains intact,
but its logical foundations can be
reordered.

The Equivalence Principle (EP) arises
naturally from three established lines
of work, rather than being a
primitive axiom.



This deck will demonstrate how these three pillars combine to support an emergent view of the EP,
leading to new, testable predictions in the quantum regime.

Pillar I: Causality Sets the Kinematic Backbone



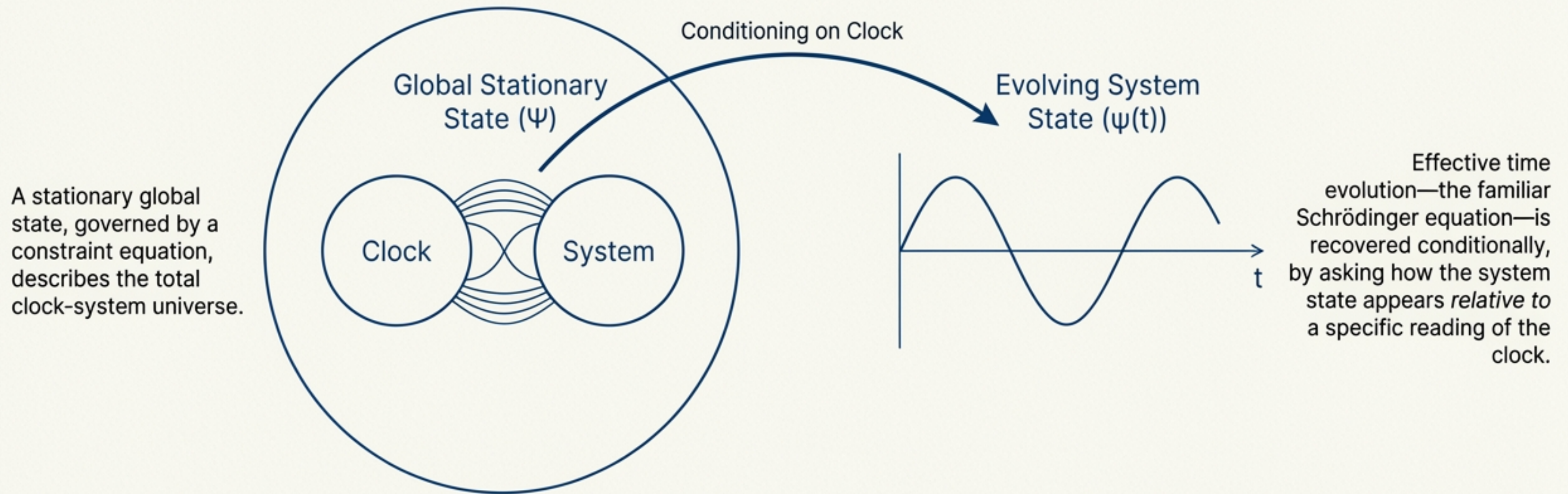
Spacetime structure can be derived operationally from minimal assumptions about observers and signals, without recourse to specific fields like electromagnetism.

- **Observers and Events:** An observer is a localized system with a clock, able to send and receive signals. Events are indexed by clock readings.
- **Finite Signaling & Distance:** If information exchange takes finite time, round-trip signal times operationally define distance. A maximal transfer speed emerges naturally.
- **Causal Order Fixes Geometry:** By Alexandrov-Zeeman type results, transformations that preserve the causal order of events necessarily form the inhomogeneous Lorentz group (up to scale).

Causality → Lorentz Group. The fundamental symmetries of special relativity are a direct consequence of causal structure. This provides the fixed inertial scaffold upon which dynamics will be built.

Pillar II: Relational Time Encodes Dynamics in Correlations

In the Page-Wootters formalism, time is not a background parameter but an emergent property of quantum correlations between a 'clock' and a 'system'.



Each choice of clock defines a valid local "temporal perspective," while the global state remains static.

A Simple Deformation Generates Universal Energy-Energy Coupling

The Mechanism

We introduce a minimal deformation to the global constraint, coupling the clock's momentum ($\hat{p}_t$) to the system's energy ($\hat{H}_S$) with a deformation scale Λ .

$$\hat{J} \equiv \hat{p}_t \otimes \hat{I}_S + \hat{I}_t \otimes \hat{H}_S + \frac{1}{\Lambda} \hat{p}_t \otimes \hat{H}_S \approx 0$$

The Result

Conditioning on the clock now yields a modified effective Hamiltonian for the system.

$$\hat{H}_{\text{eff}} = \hat{H}_S \left(\hat{I}_S + \frac{\hat{H}_S}{\Lambda} \right)^{-1}$$

Proposition 1 (Low-energy expansion)

For low system energies ($\|\hat{H}_S\|/\Lambda \ll 1$), the effective Hamiltonian expands to:

$$\hat{H}_{\text{eff}} = \hat{H}_S - \frac{1}{\Lambda} \hat{H}_S^2 + O(\Lambda^{-2})$$

Crucial Implication

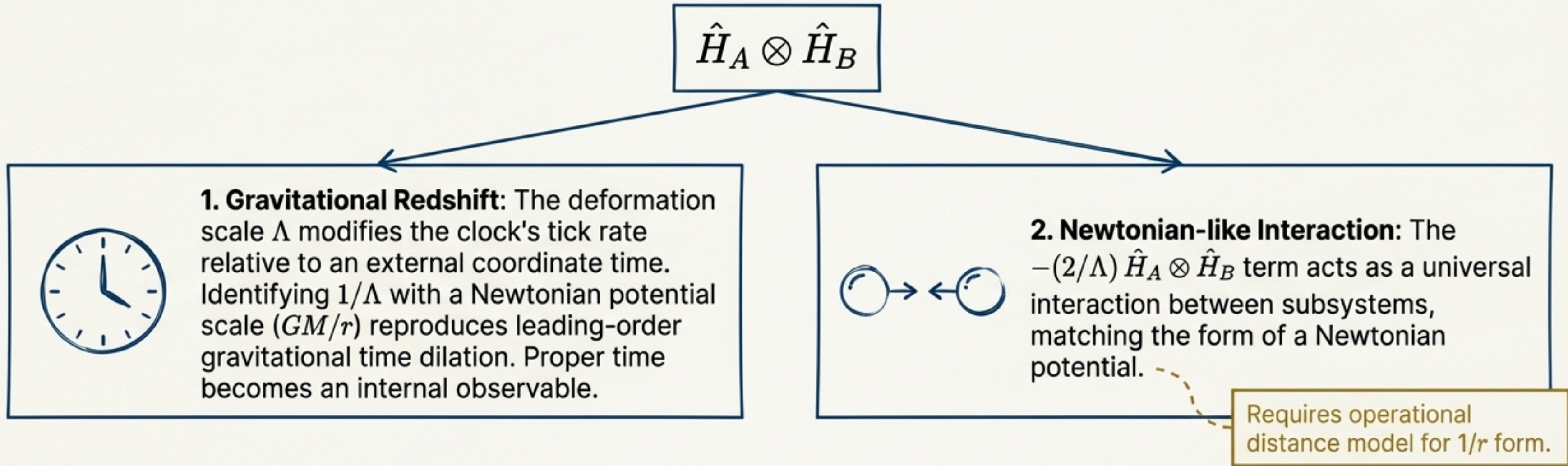
If the system is composite ($\hat{H}_S = \hat{H}_A + \hat{H}_B$), this expansion contains a universal cross-term:

$$\begin{aligned} \hat{H}_{\text{eff}} \approx & \hat{H}_A + \hat{H}_B - \frac{1}{\Lambda} (\hat{H}_A^2 + \hat{H}_B^2) + O(\Lambda^{-2}) \\ & - \frac{2}{\Lambda} \hat{H}_A \otimes \hat{H}_B \end{aligned}$$

A simple clock-system coupling in a relational framework automatically generates a universal energy-energy interaction at low energy. This term is the seed of gravitational effects.

From Energy Coupling to Redshift and Newtonian Interaction

The universal energy-energy interaction manifests as two key gravitational effects in the low-energy limit.

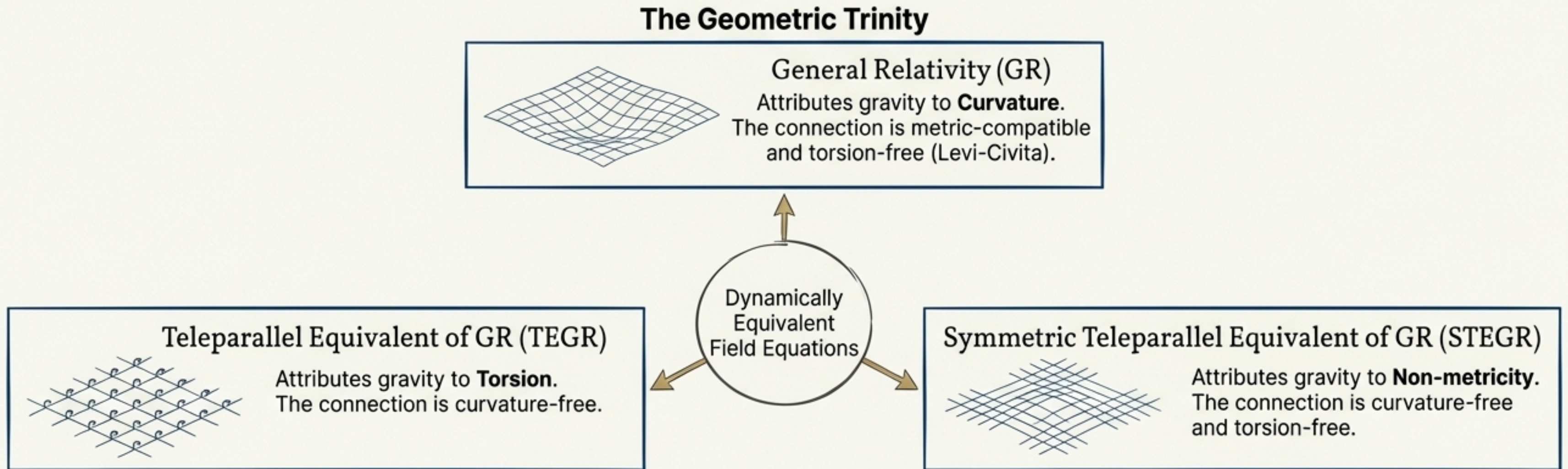


An Important Caveat: Distance Dependence

- The relational model itself does not enforce a $1/r$ spatial dependence. It directly yields the energy-energy coupling structure.
- To recover the $1/r$ law, one must import an operational definition of distance, for example, one built from signal round-trip times within the causal scaffold (from Pillar I).

Pillar III: The Geometric Trinity Shows Underdetermination

The same classical field equations can be formulated in three conceptually distinct geometric languages.
The choice of geometry is not uniquely fixed by the dynamics.



Despite their different conceptual foundations, GR, TEGR, and STEGR are dynamically equivalent (up to boundary terms).

Same predictions $\neq$ Same concepts. This equivalence shows that identifying gravity with curvature is a powerful interpretation, but not a logical necessity. Other geometric encodings are equally viable.

The Role of the Equivalence Principle Varies Across Geometries

While dynamically equivalent, the geometric trinity provides different narratives for the EP.

In Curvature-based GR

AXIOM

The EP is a foundational **axiom**. It motivates the existence of local inertial frames where gravity “disappears,” justifying the identification of gravity with spacetime geometry.

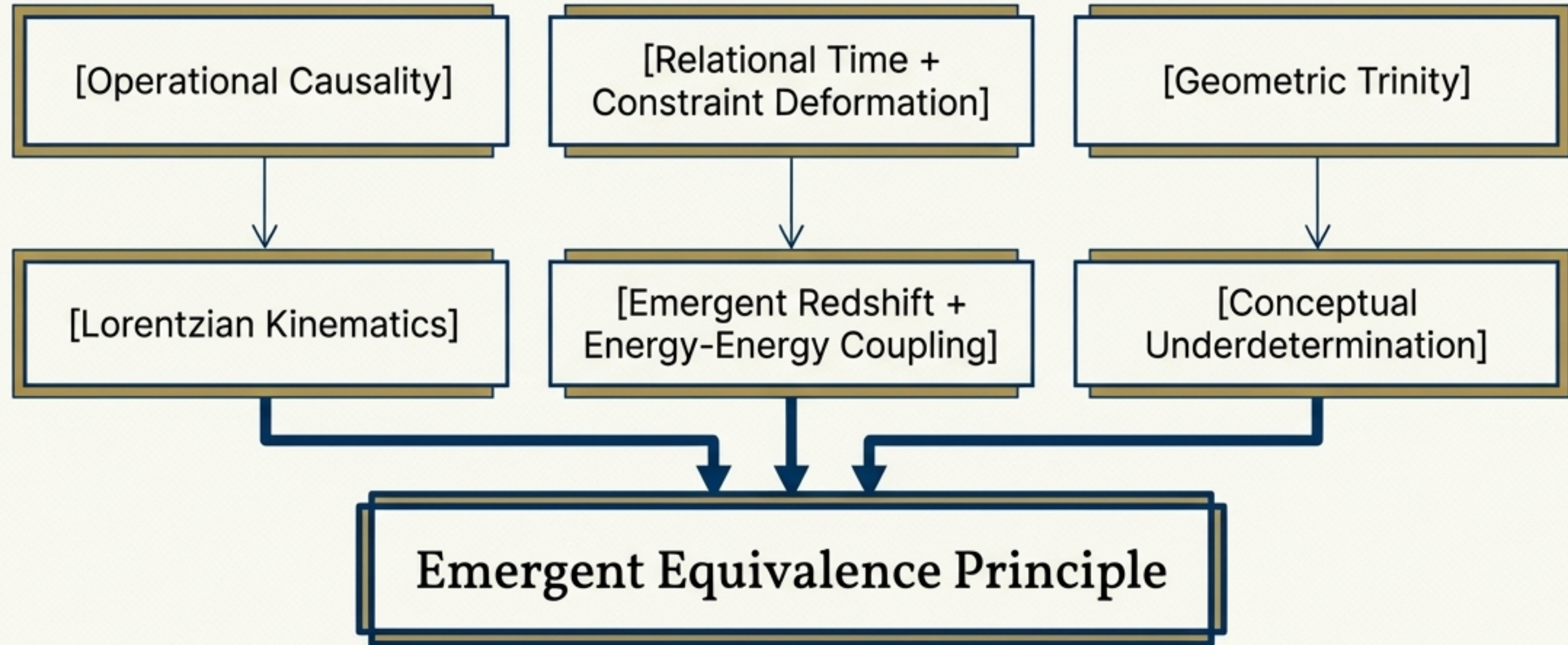
In Teleparallel Gravity (TEGR/STEGR)

CONSEQUENCE

Inertial and gravitational effects are separated at the level of the connection. EP-like phenomenology (e.g., universality of free fall) can be recovered as a **consequence** of the dynamics without being a fundamental postulate.

The existence of TEGR and STEGR provides a precedent for a framework where the EP is emergent rather than axiomatic. It decouples the tested *phenomenology* of the EP from its status as a foundational *principle*.

Synthesis: The Emergent Equivalence Principle

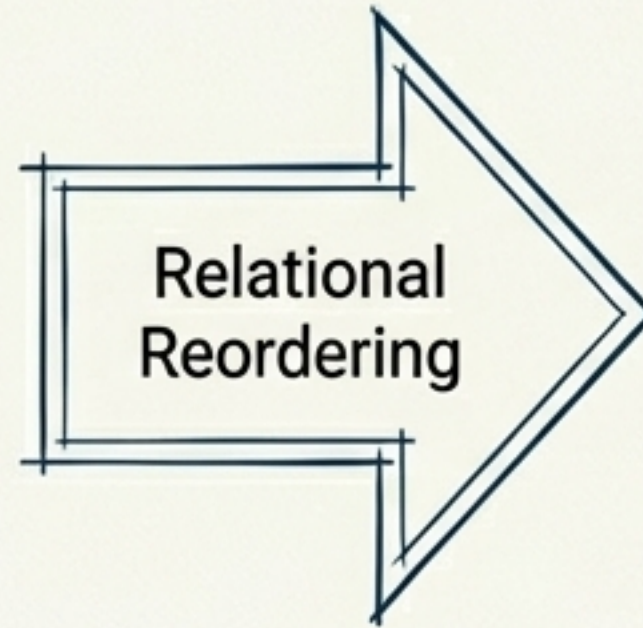


Causality fixes the Lorentz scaffold; relational time with a small deformation generates redshift and a universal energy-energy term; the geometric trinity permits multiple codings; together, these imply the EP is a low-energy symmetry with clear conditions for small deviations.

Universality as a Consequence, Not a Postulate

The Traditional View: EP as Postulates

- **Weak Equivalence Principle (WEP):**
Universality of Free Fall.
- **Einstein Equivalence Principle (EEP):**
Local physics is Lorentz invariant.
- **Strong Equivalence Principle (SEP):**
Includes gravitational self-energy.



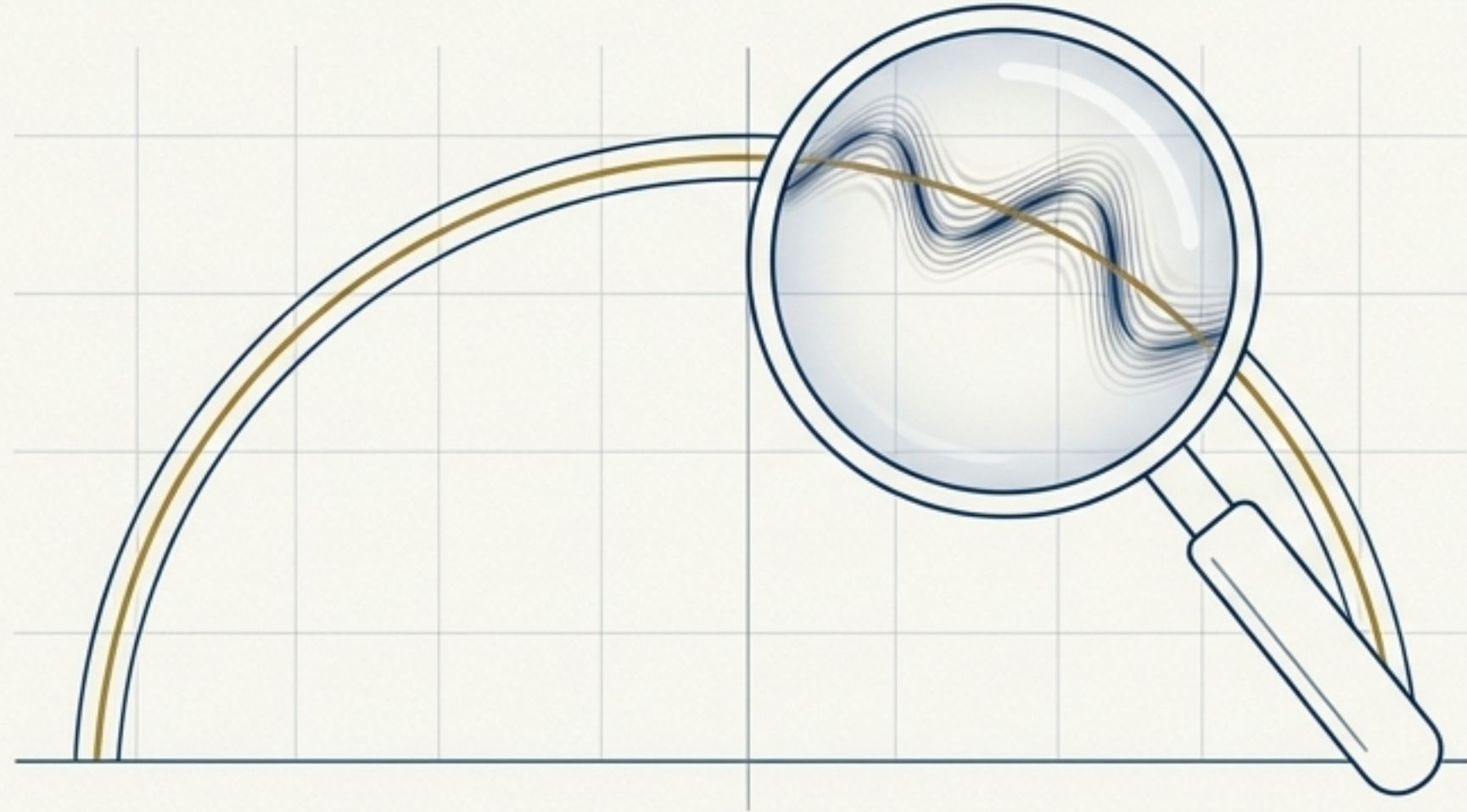
The Emergent View: EP as Consequences

- **EEP Recovery:** EEP is largely honored by Pillar I (Causality $\rightarrow$ Lorentz Group), which establishes the local inertial scaffold.
- **WEP Recovery:** WEP is recovered as a direct consequence of Pillar II. The effective coupling depends on the system's energy operator ($\hat{H}S$), not on composition-specific constants or a separate 'gravitational mass.' Therefore, the response to the interaction is universal.

The **equality of inertial and gravitational mass** is not an axiom to be assumed, but a result derived from the **universal nature** of the relational energy-energy coupling.

The New Prediction: Coherence-Dependent Deviations from the EP

Where to Look for New Physics: Deviations from the standard EP picture should not appear as classical fifth forces or composition-dependent effects. Instead, they are predicted to be subtle, quantum phenomena.



Signatures of Emergence

Deviations should appear when the low-energy approximation is challenged. They will track:

1. **Clock Resolution & Coherence:**
The degree to which a clock maintains its quantum coherence.
2. **Energy Superpositions:**
Sources or clocks existing in a superposition of different internal energy states.
3. **Perturbative Regime Departures:**
Systems where the energy $\|\hat{H}_S\|$ is not negligibly small compared to the deformation scale Λ .



These deviations manifest as **coherence-dependent corrections to conditional evolution**.
This is not a violation of the classical EP, but a quantum modification to it.

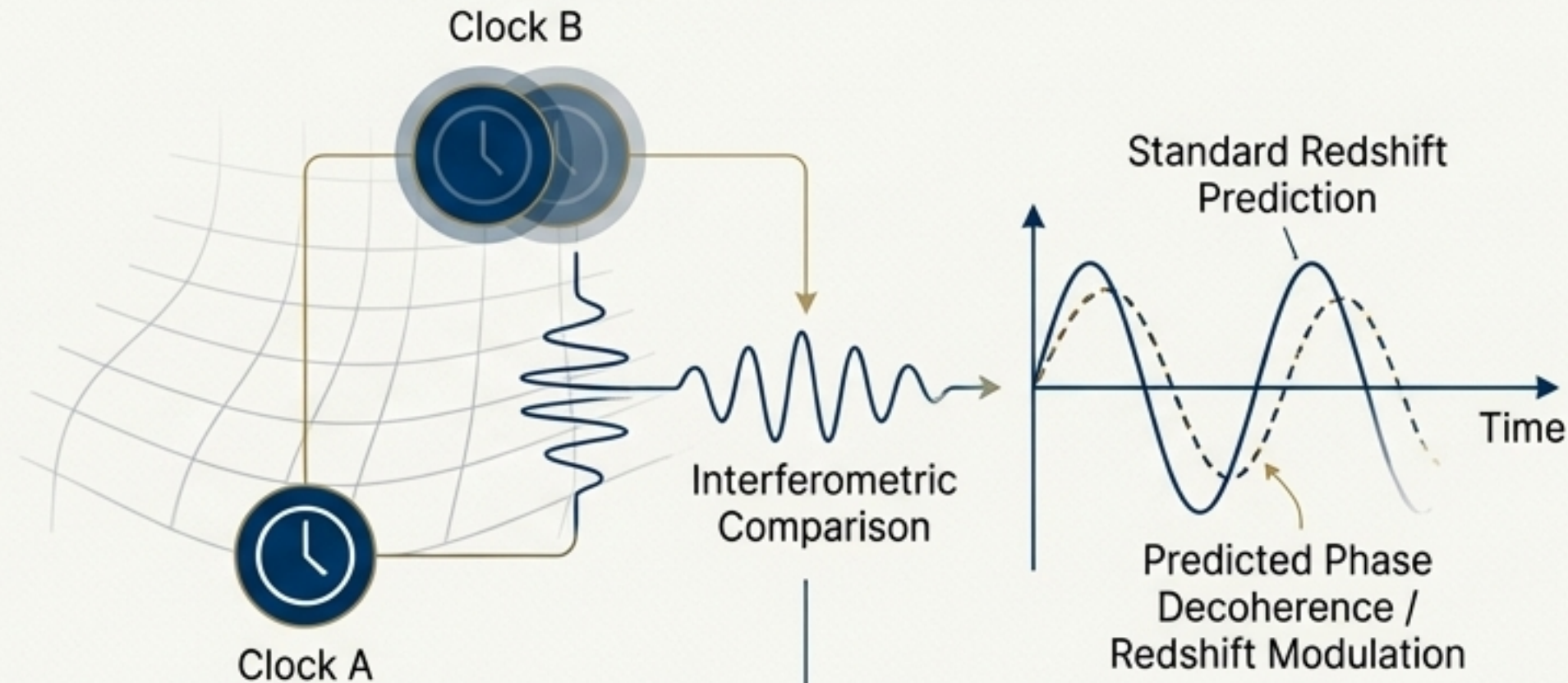
The Experimental Probe: Quantum Clock Interferometry

Quantum clock interferometry and related setups directly target the predicted coherence-dependent effects.

The Proposed Experiment

Compare two or more high-precision quantum clocks (e.g., atomic or optical lattice clocks) at different gravitational potentials.

Prepare one or both clocks in a superposition of internal energy states.



The Predicted Observable

A slight deviation from the standard gravitational redshift prediction. This deviation will manifest as a **phase decoherence** or **redshift modulation** that is correlated with the internal energy superposition of the clock states.

The Measurable Parameter

The effect is controlled by the dimensionless ratio $\varepsilon_\Lambda = \|\hat{H}S\|/\Lambda$. Modern clock sensitivities are approaching regimes where $\varepsilon_\Lambda \sim 10^{-3}$ is plausible and such effects could be detected.

Status and Outlook



What is Accounted For

- **Leading-Order Redshift:** Recovered by matching Λ^{-1} to the Newtonian potential scale.
- **Newtonian Limit:** The universal energy-energy coupling provides the correct interaction form.
- **Classical Equivalence:** The geometric trinity ensures consistency with established weak-field tests of GR.



What is an Open Problem

****The 1/r Dependence****

A principled, first-principles derivation of the inverse-square law for distance remains an open task. The current model requires importing an operational definition of distance.



The Path Forward

- **Theory:** Integrate an operational distance metric (from the causal structure) directly into the relational constraint.
- **Experiment:** Push quantum clock interferometry to probe for the predicted coherence-dependent redshift corrections.

The Equivalence Principle: From Axiom to Accessible Symmetry



The Reordered Foundation

1. **Operational Causality** fixes relativistic kinematics.
2. **Relational Dynamics** with weak deformations generate redshift and universal energy-energy interactions.
3. **The Geometric Trinity** reveals that multiple geometric descriptions can summarize this identical phenomenology.

Together, these elements motivate treating the Equivalence Principle as an emergent, low-energy symmetry. This framework positions coherence-dependent gravitational effects—testable with quantum clocks—as the primary signature of this deeper, relational structure.